M1.4 Unit Overview

Describing Functions

Students determine whether a relationship is a *function* and interpret statements in ***function notation***. They compare functions using function notation and make connections between a function's multiple representations (graphs, tables, equations, and situations). They describe functions using their key features, including ***average rate of change***, ***domain***, and ***range***. They interpret, evaluate, graph, and write equations of ***absolute value functions***. They analyze and represent sequences using function notation.

Let us know what you thought about this lesson by filling out this survey.

Big Ideas: CC2: Modeling with Functions, Comparing Models, Systems of Equations, CC3: Composing Functions

Explore: School Commutes
(Optional) Launch the unit with a non-routine task to investigate the question, "How can a graph represent a situation?"

Sub-Unit 1: Function Notation *(Lessons 1–3)*

- Determine whether a relationship is a function.
- **Language Goal:** Interpret statements in function notation.

Sub-Unit 2: Key Features of Functions *(Lessons 4–10 + Practice Day + Quiz)*

- Describe functions using their key features, including domain, range, and average rate of change.
- **Language Goal:** Compare functions using function notation and their key features.

Sub-Unit 3: Special Types of Functions *(Lessons 11–15 + Practice Day)*

- Interpret, evaluate, graph, and write equations of absolute value functions.
- **Language Goal:** Analyze and model situations using function tools, including situations that model sequences.

Contextual Vocabulary
machine, deliver, toppings, bow tie, bumper cars, Ferris wheel, roller coaster, automobile, spaceship, rental, mystery, target

Standards

Addressing A-CED.2 A-REI.11 F-BF.1 F-BF.1.a F-BF.3 F-IF.1 F-IF.2 F-IF.3 F-IF.4 F-IF.5 F-IF.6 F-IF.7 F–BF.2 F–LE.2 N-Q.1 N-Q.2 SMP.1 SMP.2 SMP.3 SMP.4 SMP.5 SMP.6 SMP.7 SMP.8

Vocabulary

- absolute value function
- average rate of change
- domain
- function notation
- interval
- maximum (of a function)
- minimum (of a function)
- negative (interval or function)
- positive (interval or function)
- range
- vertical compression
- vertical stretch

Review Vocabulary

- absolute value
- arithmetic sequence
- boundary point
- compound inequality
- constant difference
- constant ratio
- decreasing
- dependent variable
- explicit definition
- function
- geometric sequence
- increasing
- independent variable
- inequality
- input
- output
- rate of change
- recursive definition
- slope
- slope-intercept form
- translation
- y-intercept

M1.4 Pre-Unit Check

Learn more about your students' understanding of foundational concepts and skills that will support them in the upcoming unit.
This can be given in its entirety before the unit or spread throughout the unit.
This pre-unit check is intended for students to complete using a scientific calculator.

OPTIONAL This is optional and can be given in its entirety before the unit begins or spread out throughout the unit.

Let us know what you thought about this lesson by filling out this survey.

Instruction | Differentiation Beyond the Lesson

Big Ideas: CC1: Modeling with Functions, Data Explorations (Grade 8) CC2: Linear Equations (Grade 8) Multiple Representations of Functions (Grade 8), CC4: Distance and Direction (Grade 6)

Standards

Assessing 6.NS.7.c 8.EE.6 8.F.1 8.F.4 8.F.5 A-CED.2 SMP.3 SMP.4 SMP.6

Explore: School Commutes

How can a graph represent a situation?

OPTIONAL This lesson is optional as the standards are addressed in other lessons within this course.

Let us know what you thought about this lesson by filling out this survey.

Student Edition pages and **Presentation Screens** support learning in this lesson.

Big Ideas: CC2: Modeling with Functions, Comparing Models

Today's Goals

- Sketch a graph to match a situation.
- **Language Goal:** Describe how situations are represented in graphs. (*Writing, Speaking, and Listening*)

Math Language Development

Consider using the scaffolds from the *Math Language Development Resources* with the **Building Math Habits of Mind** part of this Explore to support math language acquisition for your students.

Additional Materials

coloring tools

Standards

Building On 8.F.1 8.F.5 Addressing SMP.1 SMP.2 SMP.4 Building Toward F-IF.1 F-IF.4 N-Q.1

Language Standards

ELD.PI.9-10.1 ELD.PI.9-10.10 ELD.PI.9-10.5 ELD.PI.9-10.6

Instructional Routines

- Notice and Wonder
- Discussion Supports (MLR8)

Mystery Rule

Lesson 1: What Is a Function?

Let's consider whether or not rules are functions.

Let us know what you thought about this lesson by filling out this survey.

Student devices recommended. Student Edition pages are also available.
Why digital? Animated function machines throughout the lesson support students in reasoning repeatedly with different inputs to each rule and in testing their predictions.

Instruction | Differentiation Beyond the Lesson

Big Idea: CC2: Modeling with Functions

Today's Goals

- Determine whether or not a rule is a *function*.
- **Language Goal:** Explain that a function has only one *output* for every *input*. *(Writing, Speaking, and Listening)*

Math Language Development

Consider using the scaffolds from the *Math Language Development Resources* with Activity 1 to support math language acquisition for your students.

Additional Vocabulary
relation

Standards

Building On 8.F.1 Addressing SMP.3 SMP.6 SMP.8 Building Toward F-IF.1

Language Standards

ELD.PI.9-10.1 ELD.PI.9-10.12 ELD.PI.9-10.2 ELD.PI.9-10.3 ELD.PI.9-10.7 ELD.PI.9-10.8

Review Vocabulary

- input
- output
- function

Instructional Routines

- Stronger and Clearer Each Time (MLR1)
- Collect and Display (MLR2)

Pricing Pizzas

Lesson 2: Introducing Function Notation

Let's learn what function notation is and interpret function notation statements in context.

Let us know what you thought about this lesson by filling out this survey.

Student devices recommended. Student Edition pages are also available.
Why digital? Students use the digital cash register interactions to develop conceptual understanding of function notation.

Instruction | Differentiation Beyond the Lesson

Big Idea: CC2: Modeling with Functions

Today's Goal

- **Language Goal:** Interpret a statement written in *function notation* in context. *(Reading, Writing, Speaking, and Listening)*

Math Language Development

Consider using the scaffolds from the *Math Language Development Resources* with Activity 1 to support math language acquisition for your students.

Vocabulary

- function notation

Review Vocabulary

- dependent variable
- independent variable

Instructional Routines

- Critique, Correct, Clarify (MLR3)
- Think-Pair-Share

Standards

Building On 8.F.1 Addressing F-IF.1 F-IF.2 SMP.2 SMP.3 SMP.6 Building Toward F-IF.2

Language Standards

ELD.PI.9-10.1 ELD.PI.9-10.11 ELD.PI.9-10.12 ELD.PI.9-10.2 ELD.PI.9-10.6

Toy Factory

Lesson 3: Function Notation and Equations

Let's explore functions represented as equations written in function notation.

Let us know what you thought about this lesson by filling out this survey.

Student devices recommended. Student Edition pages are also available.
Why digital? Students receive Responsive Feedback as they use interactions to try different inputs and outputs to the toy factory functions in real time.

Instruction | Differentiation Beyond the Lesson

Big Ideas: CC2: Modeling with Functions, Comparing Models

Today's Goals

- Evaluate functions written in function notation.
- Write equations of functions using function notation.
- **Language Goal:** Describe equations written in function notation in context. *(Speaking and Writing)*

Math Language Development

Consider using the scaffolds from the *Math Language Development Resources* with Activity 2 to support math language acquisition for your students.

Standards

Building On F-IF.1 Addressing A-CED.2 F-BF.1 F-IF.1 F-IF.2 SMP.7 SMP.8 Building Toward F-IF.2

Language Standards

ELD.PI.9-10.1 ELD.PI.9-10.10 ELD.PI.9-10.2 ELD.PI.9-10.3

Instructional Routines

- Notice and Wonder
- Decide and Defend

Function Carnival

Lesson 4: Creating and Interpreting Graphs of Functions

Let's create and analyze graphs that represent stories.

Let us know what you thought about this lesson by filling out this survey.

Student devices recommended. Student Edition pages are also available.
Why digital? The digital interaction generates animations of the graphs students create, supporting students in interpreting graphs and revising their graphs to match the story.

Instruction | Differentiation Beyond the Lesson

Big Ideas: CC2: Modeling with Functions, Comparing Models

Today's Goals

- Sketch a graph of a function to match a situation.
- **Language Goal:** Describe connections between situations and graphs. (*Writing, Speaking, and Listening*)

Math Language Development

Consider using the scaffolds from the *Math Language Development Resources* with Activity 1 to support math language acquisition for your students.

Review Vocabulary

- decreasing
- increasing
- rate of change

Instructional Routines

- Tell a Story
- Critique, Correct, Clarify (MLR3)

Standards

Building On 8.F.1 8.F.5 Addressing F-IF.1 F-IF.4 N-Q.2 SMP.2 SMP.4 SMP.6 Building Toward F-IF.4

Language Standards

ELD.PI.9-10.1 ELD.PI.9-10.10 ELD.PI.9-10.11 ELD.PI.9-10.12 ELD.PI.9-10.2

Craft-a-Graph

Lesson 5: Key Features of Graphs

Let's describe and create graphs of functions using key features.

This activity contains a Polygraph. Learn how to play.

Let us know what you thought about this lesson by filling out this survey.

Student devices recommended. Student Edition pages are also available.
Why digital? Repeated Challenges offer immediate Responsive Feedback to support students in developing fluency with creating functions that have specific key features.

Instruction | Differentiation Beyond the Lesson

Big Ideas: CC2: Modeling with Functions, Comparing Models

Today's Goals

- Use the key features of a function to build a graph of that function.
- **Language Goal:** Describe key features of a graph using terms like *positive*, *negative*, *maximum*, *minimum*, *increasing*, and *decreasing*. *(Reading and Writing)*

Math Language Development

Consider using the scaffolds from the *Math Language Development Resources* with Activity 2 to support math language acquisition for your students.

Standards

Building On F-IF.1 F-IF.4 Addressing F-IF.4 SMP.2 SMP.6 Building Toward F-IF.7b

Language Standards

ELD.PI.9-10.1 ELD.PI.9-10.10 ELD.PI.9-10.12 ELD.PI.9-10.5 ELD.PI.9-10.6

Vocabulary

- maximum (of a function)
- minimum (of a function)
- negative (interval or function)
- positive (interval or function)

Review Vocabulary

- decreasing (interval or function)
- increasing (interval or function)
- input
- output

Instructional Routines

- Collect and Display (MLR2)

Plane, Train, and Automobile

Lesson 6: Average Rate of Change

Let's calculate the rate of change over a specified interval.

Let us know what you thought about this lesson by filling out this survey.

Student devices recommended. Student Edition pages are also available.
Why digital? Students receive Responsive Feedback to support them in developing fluency calculating the average rate of change over various intervals and different modes of travel.

Instruction | Differentiation Beyond the Lesson

Big Idea: CC2: Modeling with Functions

Today's Goals

- Calculate the *average rate of change* over an *interval* on a graph.
- **Language Goal:** Interpret the *average rate of change* in context. *(Reading, Writing, Speaking, and Listening)*

Math Language Development

Consider using the scaffolds from the *Math Language Development Resources* with Activity 1 to support math language acquisition for your students.

Standards

Building On 8.F.4 Addressing F-IF.6 N-Q.1 SMP.2 SMP.3 Building Toward A-REI.11 F-IF.4 F-IF.6

Language Standards

ELD.PI.9-10.10 ELD.PI.9-10.12 ELD.PI.9-10.2 ELD.PI.9-10.3 ELD.PI.9-10.5 ELD.PI.9-10.6 ELD.PI.9-10.8

Vocabulary

- average rate of change
- interval

Review Vocabulary

- slope

Instructional Routines

- Tell a Story
- Notice and Wonder
- Compare and Connect (MLR7)
- Collect and Display (MLR2)

Space Race

Lesson 7: Comparing Graphs

Let's make connections between function notation and key features of graphs.

Let us know what you thought about this lesson by filling out this survey.

Student devices recommended. Student Edition pages are also available.
Why digital? Students receive Responsive Feedback as they use statements written in function notation to compare functions.

Instruction | Differentiation Beyond the Lesson

Big Ideas: CC2: Systems of Equations, Modeling with Functions

Today's Goals

- Interpret statements written in function notation and use them to compare key features of graphs.
- **Language Goal:** Compare and contrast key features of functions using statements written in function notation. *(Reading, Writing, Speaking, and Listening)*

Math Language Development

Consider using the scaffolds from the *Math Language Development Resources* with Activity 1 to support math language acquisition for your students.

Standards

Building On 8.F.5 Addressing A-REI.11 F-IF.2 F-IF.4 F-IF.6 SMP.6 Building Toward A-REI.11 F-IF.9

Language Standards

ELD.PI.9-10.1 ELD.PI.9-10.2 ELD.PI.9-10.3 ELD.PI.9-10.5

Instructional Routines

- Compare and Connect (MLR7)
- Think-Pair-Share

Ins and Outs

Lesson 8: Introducing Domain and Range

Let's explore the possible inputs and outputs of functions.

Let us know what you thought about this lesson by filling out this survey.

Student Edition pages and **Presentation Screens** support learning in this lesson.

Instruction | Differentiation Beyond the Lesson

Big Idea: CC2: Modeling with Functions

Today's Goals

- Determine if a value is a possible input or output for a function given in context.
- **Language Goal:** Describe the *domain* and *range* of a function as the set of all possible inputs and outputs. *(Writing and Speaking)*

Math Language Development

Consider using the scaffolds from the *Math Language Development Resources* with Activity 2 to support math language acquisition for your students.

Additional Vocabulary

discrete, continuous

Addition Material

Show What You Know Sheet

Standards

Building On 8.F.1 Addressing F-IF.1 F-IF.5 N-Q.2 SMP.3 SMP.6 Building Toward F-IF.5 F-IF.7 F-IF.7b

Language Standards

ELD.PI.9-10.1 ELD.PI.9-10.10 ELD.PI.9-10.12 ELD.PI.9-10.2 ELD.PI.9-10.3 ELD.PI.9-10.6

Vocabulary

- domain
- range

Review Vocabulary

- dependent variable
- function
- independent variable

Instructional Routines

- Which One Doesn't Belong?
- Think-Pair-Share

Elevator Stories

Lesson 9: Describing Domain and Range With Inequalities

Let's use compound inequalities to describe the domain and range of functions from their graphs.

Let us know what you thought about this lesson by filling out this survey.

Student devices recommended. Student Edition pages are also available.
Why digital? Students use digital interactions to make connections between situations, tables, and graphs.

Instruction | Differentiation Beyond the Lesson

Big Idea: CC2: Modeling with Functions

Today's Goals

- Write the domain and range of a function using inequalities.
- **Language Goal:** Interpret the meaning of the domain and range in context. *(Writing, Speaking, and Listening)*

Math Language Development

Consider using the scaffolds from the *Math Language Development Resources* with Activity 2 to support math language acquisition for your students.

Review Vocabulary

- compound inequality

Instructional Routines

- Tell a Story
- Decide and Defend
- Compare and Connect (MLR7)

Standards

Building On F-IF.2 F-IF.4 Addressing F-IF.5 SMP.1 SMP.2 SMP.3 SMP.6 Building Toward F-IF.7b N-Q.2

Language Standards

ELD.PI.9-10.1 ELD.PI.9-10.12 ELD.PI.9-10.3 ELD.PI.9-10.4 ELD.PI.9-10.5

Marbleslides

Lesson 10: Graphing Functions With Restrictions

Let's practice restricting the domain and range of a graph.

Let us know what you thought about this lesson by filling out this survey.

Student devices recommended. Student Edition pages are also available.
Why digital? Students receive Responsive Feedback as they develop fluency with restricting the domain or range of a graph while playing Marbleslides.

Instruction | Differentiation Beyond the Lesson

Big Idea: CC2: Modeling with Functions

Today's Goal

- Restrict the domain and range of a function using inequalities.
- **Language Goal:** Explain how to restrict the domain and range of a function using inequalities. (*Speaking*)

Math Language Development

Consider using the scaffolds from the *Math Language Development Resources* with Activity 1 to support math language acquisition for your students.

Additional Vocabulary

restriction

Standards

Building On F-IF.1 Addressing F-IF.5 SMP.1 SMP.6 Building Toward F-BF.3 F-IF.7b

Language Standards

ELD.PI.9-10.1 ELD.PI.9-10.2 ELD.PI.9-10.3

Review Vocabulary

- slope
- y-intercept
- slope-intercept form
- compound inequality

Instructional Routines

- Decide and Defend
- Discussion Supports (MLR8)

M1.4 Practice Day 1

Students practice the concepts, skills, and strategies developed in Lessons 1–10. Consider using this Practice Day as preparation for the upcoming Sub-Unit Quiz.

Let us know what you thought about this lesson by filling out this survey.

Big Ideas: CC2: Modeling with Functions, Comparing Models, Systems of Equations

Additional Materials

Option 1: Stations

- Task cards (two sets for the whole class)

Option 2: Level Up

- Task cards (one set per group)

Standards

Addressing A-REI.11 F-IF.2 F-IF.4 F-IF.5 F-IF.6 SMP.2 SMP.6

M1.4 Sub-Unit Quiz

Learn about your students' understanding of the concepts and skills so far in this unit.
This assessment is intended for students to complete using a scientific calculator.

Let us know what you thought about this lesson by filling out this survey.

Instruction | Differentiation Beyond the Lesson

Big Ideas: CC2: Modeling with Functions, Comparing Models

Standards

Assessing F-BF.1 F-IF.1 F-IF.2 F-IF.4 F-IF.5 F-IF.6 SMP.2 SMP.4 SMP.6

Recursion Excursion

Lesson 11: Writing Sequences in Function Notation

Let's write a recursive definition for a sequence using function notation.

Let us know what you thought about this lesson by filling out this survey.

Student devices recommended. Student Edition pages are also available.
Why digital? Students use a digital interaction they are familiar with to apply what they know about arithmetic and geometric sequences.

Instruction   Differentiation Beyond the Lesson

Big Idea: CC3: Composing Functions

Today's Goals

- Write a recursive definition for a sequence using function notation.
- **Language Goal:** Describe how a sequence is a function with a domain that is a subset of the integers. *(Speaking and Listening)*

Math Language Development

Consider using the scaffolds from the *Math Language Development Resources* with Activity 1 to support math language acquisition for your students.

Standards

Building On F-BF.1 F-IF.2 F-LE.2 Addressing F-BF.1.a F-IF.3 F-BF.2 SMP.1 SMP.2 SMP.8 Building Toward F-BF.2

Language Standards

ELD.PI.9-10.1 ELD.PI.9-10.11 ELD.PI.9-10.3 ELD.PI.9-10.5

Review Vocabulary

- arithmetic sequence
- constant difference
- recursive definition
- geometric sequence
- constant ratio

Instructional Routines

- Notice and Wonder
- Compare and Connect (MLR7)
- Critique, Correct, Clarify (MLR3)

Functions and Sequences

Lesson 12: Using Functions to Model Sequences

Let's look at different ways to write sequences.

Let us know what you thought about this lesson by filling out this survey.

Student Edition pages and an **assignable Digital Companion** needed.

Instruction | Differentiation Beyond the Lesson

Big Ideas: CC2: Modeling with Functions, CC3: Composing Functions

Today's Goals

- Write recursive and explicit definitions for sequences using function notation.
- **Language Goal:** Compare and contrast the recursive definition and the explicit definition of a sequence written in function notation. *(Speaking and Listening)*

Math Language Development

Consider using the scaffolds from the *Math Language Development Resources* with Activity 1 to support math language acquisition for your students.

Additional Materials

Activity 2 Cards (one set per pair, pre-cut); Show What You Know Sheet

Standards

Building On F-BF.1 F-IF.2 Addressing F-BF.1.a F-IF.3 F–BF.2 F–LE.2 SMP.5 SMP.7

Language Standards

ELD.PI.9-10.1 ELD.PI.9-10.11 ELD.PI.9-10.3

Review Vocabulary

- constant difference
- constant ratio
- explicit definition
- recursive definition

Instructional Routines

- Decide and Defend
- Think-Pair-Share
- Discussion Supports (MLR8)

What's Your Score?

Lesson 13: Absolute Value Functions, Part 1

Let's make sense of absolute value functions.

Let us know what you thought about this lesson by filling out this survey.

Student devices recommended. Student Edition pages are also available.

Why digital? Students use digital interactions to develop conceptual understanding of the meanings, equation and graph of absolute value functions while playing a fun game.

Instruction | Differentiation Beyond the Lesson

Big Idea: CC2: Modeling with Functions

Today's Goals

- Evaluate and interpret outputs for *absolute value* functions.
- **Language Goal:** Describe how the absolute value function $f(x) = |x - h|$ is related to the distance from a number. *(Writing and Speaking)*

Math Language Development

Consider using the scaffolds from the *Math Language Development Resources* with Activity 2 to support math language acquisition for your students.

Vocabulary

- absolute value function

Review Vocabulary

- absolute value

Instructional Routines

- Which One Doesn't Belong?
- Decide and Defend
- Discussion Supports (MLR8)

Standards

Building On 6.NS.7.c F-IF.1 Addressing F-IF.2 F-IF.7 SMP.2 SMP.7 Building Toward F-IF.7b

Language Standards

ELD.PI.9-10.1 ELD.PI.9-10.10 ELD.PI.9-10.11 ELD.PI.9-10.12 ELD.PI.9-10.3 ELD.PI.9-10.6

Absolute Value Machines

Lesson 14: Absolute Value Functions, Part 2

Let's graph absolute value functions.

Let us know what you thought about this lesson by filling out this survey.

Student devices recommended. Student Edition pages are also available.
Why digital? Students use Responsive Feedback to make connections between absolute value functions, tables, and graphs and build fluency in graphing absolute value functions.

Instruction | Differentiation Beyond the Lesson

Big Ideas: CC2: Modeling with Functions, Comparing Models, CC3: Composing Functions

Today's Goals

- Graph an absolute value function.
- **Language Goal:** Analyze the key features (e.g., minimum, stretch or compression) of an absolute value function. *(Reading, Writing, Speaking, and Listening)*

Math Language Development

Consider using the scaffolds from the *Math Language Development Resources* with Activity 3 to support math language acquisition for your students.

Vocabulary

- vertical compression
- vertical stretch

Review Vocabulary

- translation

Instructional Routines

- Notice and Wonder
- Compare and Connect (MLR7)
- Decide and Defend

Standards

Building On F-IF.1 F-IF.2 Addressing A-CED.2 F-BF.3 F-IF.7 SMP.3 SMP.6 SMP.7 Building Toward F-BF.3 F-IF.7b

Language Standards

ELD.PI.9-10.1 ELD.PI.9-10.10 ELD.PI.9-10.11 ELD.PI.9-10.12 ELD.PI.9-10.3 ELD.PI.9-10.6

Our Math Stories

Lesson 15: Using Functions to Tell Stories

Let's use functions and graphs to model math stories.

Let us know what you thought about this lesson by filling out this survey.

Student Edition pages and **Presentation Screens** support learning in this lesson.

Instruction | Differentiation Beyond the Lesson

Big Idea: CC2: Modeling with Functions, Comparing Models

Today's Goals

- Create graphs of functions that represent real-life relationships.
- **Language Goal:** Describe and interpret real-life relationships using function tools learned throughout this unit. *(Reading, Writing, Speaking, and Listening)*

Math Language Development

Consider using the scaffolds from the *Math Language Development Resources* with Activity 1 to support math language acquisition for your students.

Additional Materials

Desmos Graphing Calculator (as needed), Activity 2 Sheet, Show What You Know Sheet

Standards

Building On F-IF.1 F-IF.3 Addressing F-IF.2 F-IF.4 F-IF.5 N-Q.1 SMP.2 SMP.4 SMP.6 Building Toward F-BF.3 F-IF.9 N-Q.2 N-Q.3

Language Standards

ELD.PI.9-10.1 ELD.PI.9-10.10 ELD.PI.9-10.5

Instructional Routines

- Notice and Wonder

M1.4 Practice Day 2

Students practice the concepts, skills, and strategies developed in Lessons 1–15. Consider using this Practice Day as preparation for the upcoming End-of-Unit Assessment

Let us know what you thought about this lesson by filling out this survey.

Big Ideas: CC2: Modeling with Functions, Comparing Models, CC3: Composing Functions

Materials

Option 1: Solve and Swap

- Problem Cards (one copy per eight students, pre-cut)

Option 2: Partner Practice

- Problem Cards (one set per pair, pre-cut)

Standards

Addressing F-BF.1.a F-BF.3 F-IF.2 F-IF.4 F-IF.5 F-IF.6 SMP.2 SMP.3 SMP.6

Language Standards

ELD.PI.9-10.2

M1.4 Practice Day 2

Students practice the concepts, skills, and strategies developed in Lessons 1–15. Consider using this Practice Day as preparation for the upcoming End-of-Unit Assessment

Let us know what you thought about this lesson by filling out this survey.

Big Ideas: CC2: Modeling with Functions, Comparing Models, CC3: Composing Functions

Materials

Option 1: Solve and Swap

- Problem Cards (one copy per eight students, pre-cut)

Option 2: Partner Practice

- Problem Cards (one set per pair, pre-cut)

Standards

Addressing F-BF.1.a F-BF.3 F-IF.2 F-IF.4 F-IF.5 F-IF.6 SMP.2 SMP.3 SMP.6

Language Standards

ELD.PI.9-10.2

M1.4 Practice Day 2

Students practice the concepts, skills, and strategies developed in Lessons 1–15. Consider using this Practice Day as preparation for the upcoming End-of-Unit Assessment

Let us know what you thought about this lesson by filling out this survey.

Big Ideas: CC2: Modeling with Functions, Comparing Models, CC3: Composing Functions

Materials

Option 1: Solve and Swap

- Problem Cards (one copy per eight students, pre-cut)

Option 2: Partner Practice

- Problem Cards (one set per pair, pre-cut)

Standards

Addressing F-BF.1.a F-BF.3 F-IF.2 F-IF.4 F-IF.5 F-IF.6 SMP.2 SMP.3 SMP.6

Language Standards

ELD.PI.9-10.2

M1.4 Performance Task

Assign this summative assessment performance task at the end of the unit to evaluate students' proficiency with the concepts and skills addressed in the unit.

Let us know what you thought about this lesson by filling out this survey.

Big Ideas: CC2: Modeling with Functions, Comparing Models, CC3: Composing Functions

Standards

Assessing F-BF.3 F-IF.2 F-IF.4 F-IF.5 F-IF.7 SMP.2 SMP.3 SMP.4 SMP.6

Unit Synthesis and Reflection

Here are six optional activities for students to engage in at the end of a unit to synthesize and/or reflect on their learning from the unit.

Notes:

- **Select one to two tasks** for students to complete before or after the End-Unit Assessment.
- Many tasks invite students to refer to the Student Goals and Glossary, which can be downloaded from the Unit Overview page at the beginning of each unit.

Let us know what you thought about this lesson by filling out this survey.

Learning Goals

- Be purposeful, strategic, and goal-oriented about learning.
- Connect different ideas from the unit and across units.
- Revisit ideas from the unit in order to make revisions and reflect on their growth.